

Dictionary = It is used to store data in key value pairs

- mutable
- unordered
- no duplicates values

```
In [1]: dict = {  
        "name" : "yasir",  
        "age" : 20,  
        "cgpa" : 7.7  
    }  
  
    print(dict)
```

```
{'name': 'yasir', 'age': 20, 'cgpa': 7.7}
```

```
In [2]: print(dict.get("name"))  
        print(dict.get("age"))  
        print(dict.get("cgpa"))
```

```
yasir  
20  
7.7
```

```
In [3]: print(dict.keys())
```

```
dict_keys(['name', 'age', 'cgpa'])
```

```
In [4]: print(dict.values())
```

```
dict_values(['yasir', 20, 7.7])
```

```
In [5]: print(dict.items())
```

```
dict_items([('name', 'yasir'), ('age', 20), ('cgpa', 7.7)])
```

```
In [6]: print(dict.update({"age": "21"}))  
        print(dict)
```

```
None  
{'name': 'yasir', 'age': '21', 'cgpa': 7.7}
```

```
In [7]: print(dict.pop("name"))
```

```
yasir
```

```
In [8]: print("age" in dict)  
        print("number" in dict)
```

```
True  
False
```

```
In [9]: print(len(dict))
```

```
2
```

```
In [10]: dict["my_age"] = 20  
         print(dict)
```

```
{'age': '21', 'cgpa': 7.7, 'my_age': 20}
```

```
In [11]: print(dict["age"])
```

21

```
In [24]: dict_new = {
    "name" : "yasir",
    "age" : 20,
    "cgpa" : 7.7
}

del dict_new["age"]
print(dict_new)

{'name': 'yasir', 'cgpa': 7.7}
```

```
In [13]: # Check if a key exists in the dictionary.
d = {
    "name" : "yasir",
    "age" : 20,
    "cgpa" : 7.7
}

print("age" in d)
```

True

```
In [14]: # Merge two dictionaries.

dict1 = {'a':1,'b':2}
dict2 = {'c':3,'d':4}

dict1.update(dict2)
print(dict1)

{'a': 1, 'b': 2, 'c': 3, 'd': 4}
```

```
In [15]: # Iterate through a dictionary and print keys and values

d = {
    "name" : "yasir",
    "age" : 20,
    "cgpa" : 7.7
}

for key, values in d.items():
    print(f"key : {key}, values : {values}")
```

key : name, values : yasir

key : age, values : 20

key : cgpa, values : 7.7

```
In [16]: # Use .get() method with a default value.

d = {
    "name" : "yasir",
    "age" : 20,
    "cgpa" : 7.7
```

```
}  
  
x = d.get('city' , 'unknown')  
print(x)
```

unknown

```
In [27]: # Create a dictionary from two lists using zip().  
key = ["name", "age"]  
values = ["yasir", 20]  
del dict  
my_dict = dict(zip(key, values))  
print(my_dict)
```

{'name': 'yasir', 'age': 20}

```
In [18]: subject = ["maths","python"]  
marks = [74,75]  
  
for subject , marks in zip(subject,marks):  
    print(subject, "=", marks)
```

maths = 74
python = 75

In []: